

Asher Siddiqui

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EXPERIENCE

Arclab Solutions

Junior Software Engineer | TypeScript, React, Go, Kafka, SpiceDB, PostgreSQL, Redis

Remote

July. 2025 – Present

- Built an internal React dashboard for configuring and monitoring simulated ASIC miner fleets, consuming a Go WebSocket backend for real time fleet state.
- Architected an internal resource scoped authorization layer using SpiceDB, enabling secure multi-tenant access control across core platform APIs.
- Developed a FastAPI data export service for TimescaleDB metrics, enabling downstream reporting pipelines to consume structured time series data over a REST interface.
- Strengthened internal API security by rolling out mTLS service-to-service protection and a TLS-capable HTTP client, closing trust gaps between services.
- Secured event ingestion by adding Kafka SASL/SCRAM support and updating Docker Compose to provision secure broker users, improving production parity.
- Authored PostgreSQL/TimescaleDB migrations and performance indexes across multiple product domains.

TAFSOL Technologies

TypeScript, React, Go, MySQL, WebSockets, Docker, AWS

Karachi, Pakistan (Hybrid)

July. 2023 – July. 2025

- Built a full-stack real-time chat application with a React frontend and a Go/WebSocket backend; reduced message delivery latency from 1.3s to 0.4s via server-side LRU caching.
- Optimized authentication flow through concurrent query processing and request batching, cutting response time from 7s to 1.5s.
- Designed a custom RBAC middleware with per-route permission enforcement, integrated across REST endpoints and frontend route guards.
- Built a QR-based quick pay feature with a React UI and REST backend for internal payments, reducing transaction steps by 40%.

Open Source Contributions

Typescript, Node.js, PostgreSQL

- Refactored GitHub OAuth architecture to a database backed source of truth, eliminating fragmented token state, resulting in a cleaner data flow, and more readable codebase. ([PR #1](#), [PR #2](#))
- Designed a provider agnostic context system across Claude, Gemini, Codex, and OpenCode, reconstructing session state on provider switches and branch forks. ([PR #1](#), [PR #2](#), [PR #3](#))
- Built an agentic skills lifecycle system, handling consistency across sandboxes, and global prompt injection to ensure skill awareness across all providers. ([PR #1](#), [PR #2](#))

PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | Kubernetes, Ansible, Prometheus, Falco, Go, Python | [GitHub](#)

- Deployed a bare metal Kubernetes cluster and a botnet using Ansible to create simulated DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline with Falco and Node Exporter to collect system calls and resource metrics, ensuring centralized logging and data generation.
- Developed and deployed attack scripts to simulate and orchestrate attacks via a C2 server and botnet, enabling automated DDoS attacks.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL

Frontend: React, Next.js, WebSockets

Backend: Node.js, Express, FastAPI, Echo (Go), REST APIs

Databases: PostgreSQL, MySQL, Redis, SpiceDB

Infrastructure: Docker, Docker Compose, Kubernetes, AWS (EC2, S3), Linux, Kafka

EDUCATION

FAST - National University of Computer and Emerging Sciences (NUCES)

Bachelor of Science in Software Engineering

Karachi, Pakistan

August 2021 – June 2025

Awards & Honors: Dean's List of Honors (3.5+ GPA) in Fall 2023, Spring 2024, Fall 2024